Chat Log 2

Transcript

1. **User prompt:**

    - Provided instruction files and required that I read/follow them before making changes.
    - Included rule: do not create explanation files unless directly requested.

    - Acknowledged and followed those instructions for all subsequent work.

2. **User prompt:**

    - Provided environment/workspace details (Windows, PowerShell, project structure).

    - Used that context for file paths and command/tool behavior.

3. **User prompt:**

    - Added subagent instruction: always use run_subagent when a matching agent exists (available: Plan).

    - Delegated discovery planning to the Plan subagent before investigating code.

4. **User prompt:**

    - "Search through this project and answer this question: How does the prediction system determine if the buy / short prediction was a success. Is it just if the next data point was higher or lower? Show where in the code this is defined"

    - Ran run_subagent (Plan) to map likely files/symbols.
    - Searched and read relevant files:
        - src/kvant/labeling/triple_barrier.py
        - src/kvant/training/metrics.py
        - src/kvant/evaluation/runner.py
        - src/kvant/experiment/prepare.py
        - tests/test_triple_barrier_labeling.py

    - Replied that success is **not** based on only the next data point.
    - Explained that labels come from triple-barrier logic over a future window (TP/SL/time exit), and buy/short correctness is measured by prediction matching those labels (y_true == y_pred / m["label"] == yp).

5. **User prompt:**

    - "When are the barriers?"

    - Explained barrier timing/placement:

- Price barriers set at entry open: entry * (1 +/- height).
- Vertical barrier at entry_ts + width_minutes.
- Hits checked on future bars only (t+1 onward).

6. **User prompt:**

   - "so tb_L20_w20_h0.5_TBPD195_VS0.5-2 means 20 seconds between each barrier (w20)? What does the other parts of prepared data mean"

   - Queried naming source in code and read experiment artifacts:
     - src/kvant/experiment/prepare.py
     - prepared/tb_L20_w20_h0.5_TBPD195_VS0.5-2/config.json
     - prepared/last_experiment.txt
     - prepared/.../density_summary.json
     - prepared/.../sampler_global_meta.json
     - prepared/.../sampler_per_ticker_meta.json

   - Clarified that w20 is **20 minutes**, not seconds.
   - Decoded experiment name parts:
     - L20: lookback length 20
     - w20: width/vertical horizon 20 minutes
     - h0.5: 0.5% barrier half-height
     - TBPD195: target bars/day 195
     - VS0.5-2: volatility scaling clamp [0.5, 2.0]

   - Explained core files in the prepared folder.

7. **User prompt:**

   - "What does the vertical horizon mean in this case?"

   - Explained vertical horizon as maximum holding time/deadline for label resolution.
   - For w20 on 1m bars: about 20 future bars.
   - If no TP/SL hit before deadline, label becomes time-exit (1).

8. **User prompt:**

   - "Does lookback and width minutes have to match then?"

   - Replied: no, they are independent knobs.
   - lookback_L controls input history length.
   - width_minutes controls target/label future horizon.

9. **User prompt:**

   - "What is meant by medium-horizon"

   - Explained horizon as how far ahead labels look.
   - Positioned w20 (on 1m bars) as roughly a 20-minute, medium near-term horizon.

10. **User prompt (current):**
    - "Write this entire chat log into chat-logs-2.md. It contain all my prompts and what you did / replied."
    - Wrote this transcript into ignore/chat-logs-2.md.